

Entropy-Aware Model Predictive Control for *Transient Quality Mitigation* in Wire-Arc Additive Manufacturing

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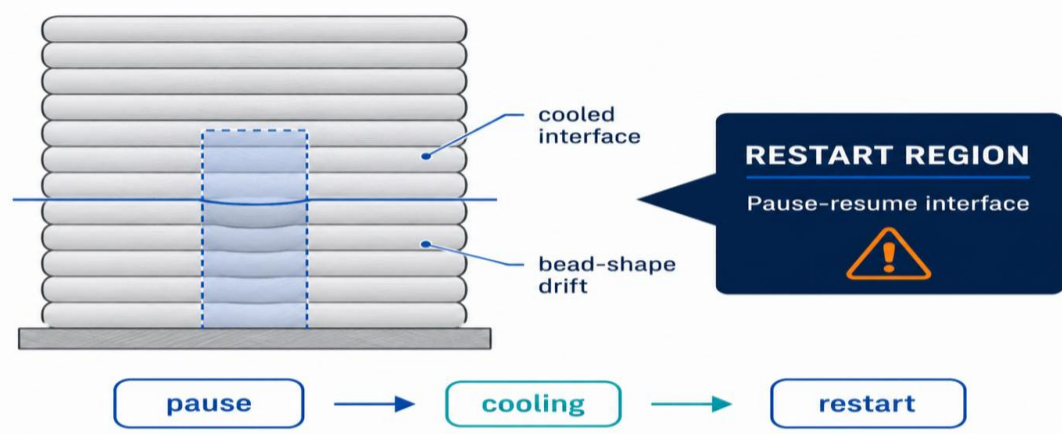


- 1 Cold start
Heat sinks after pause
- 2 Entropy collapse
Arc sound shifts sharply
- 3 Defect risk
Quality degrades post-restart
- 4 Entropy-aware MPC
Entropy guides heat correction
- 5 Front-loaded heat
Short pulse restores stability

01 / MOTIVATION

Restarts can look electrically normal while quality silently degrades.

Pause-induced cooling creates a **restart transient**; current and voltage may stay nominal, but bead morphology and quality can drift.



CURRENT
Nominal

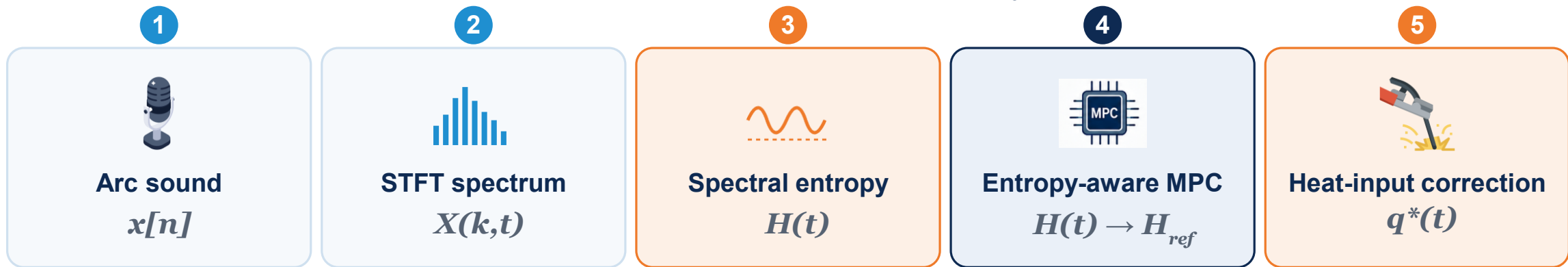
VOLTAGE
Nominal

QUALITY
At risk

02 / ENTROPY-AWARE MPC ARCHITECTURE

Arc sound becomes a control-ready entropy state.

Spectral entropy converts restart sound into $H(t)$; MPC penalizes deviation from H_{ref} and computes the corrective heat input $q^*(t)$.



ACOUSTIC SPECTRAL ENTROPY

$$H(t) = -\sum_{k=1}^K p_k(t) \log p_k(t)$$

ENTROPY-AWARE MPC OBJECTIVE

$$q^*(t) = \arg \min_q [(H - H_{ref})^2 + \lambda(\Delta q)^2]$$

Acoustic entropy is the response-side risk signal; heat input is the corrective control action.

03 / ABLATION SCOREBOARD

Entropy objective drives the performance gain.

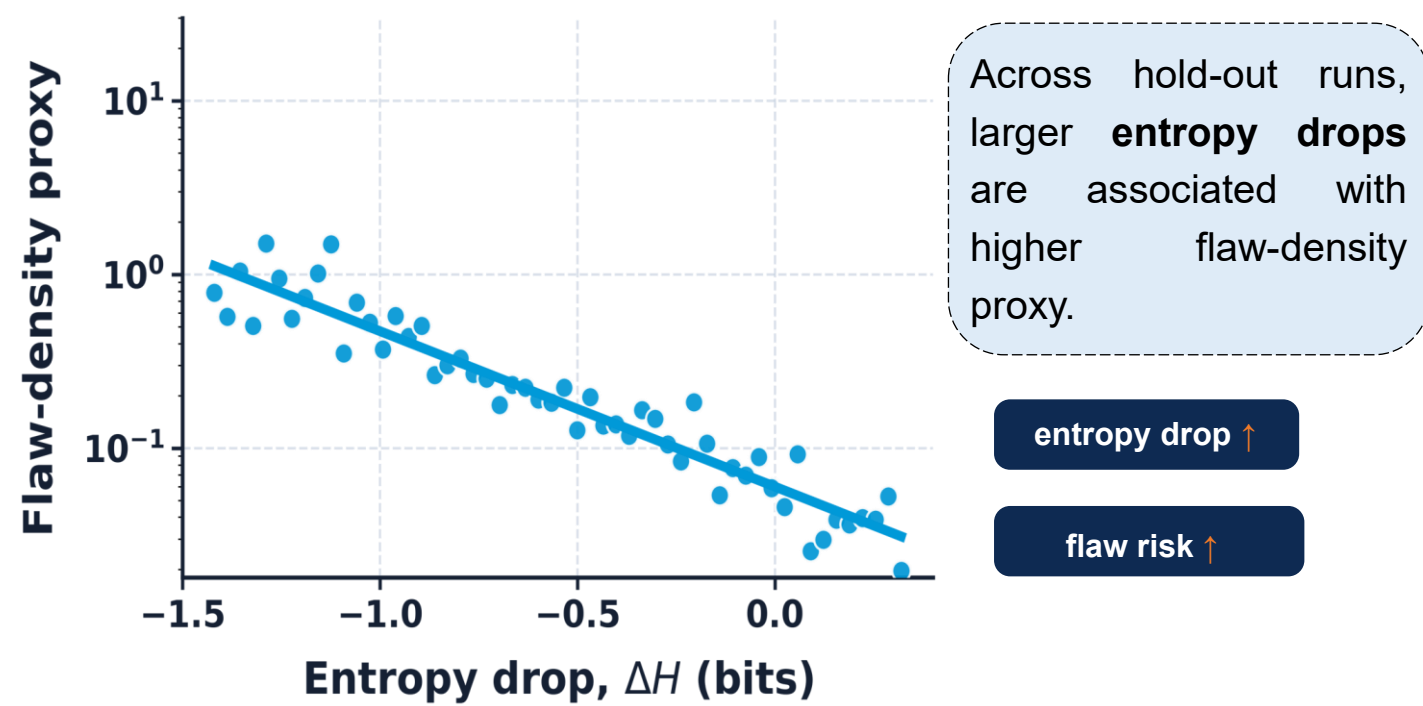
Adding entropy to MPC reduces **settling time** and **entropy-risk AUC** across both DMDc and SINDy models.

CONTROLLER	SETTLING TIME (S)	RISK AUC
M1 DMDc + Thermal	6.7	0.222
M2 DMDc + Entropy-MPC	2.8	0.054
M3 SINDy + Thermal	5.1	0.188
M4 SINDy + Entropy-MPC	3.6	0.145

★ Best: DMDc + Entropy-MPC

04 / SIGNAL VALIDATION

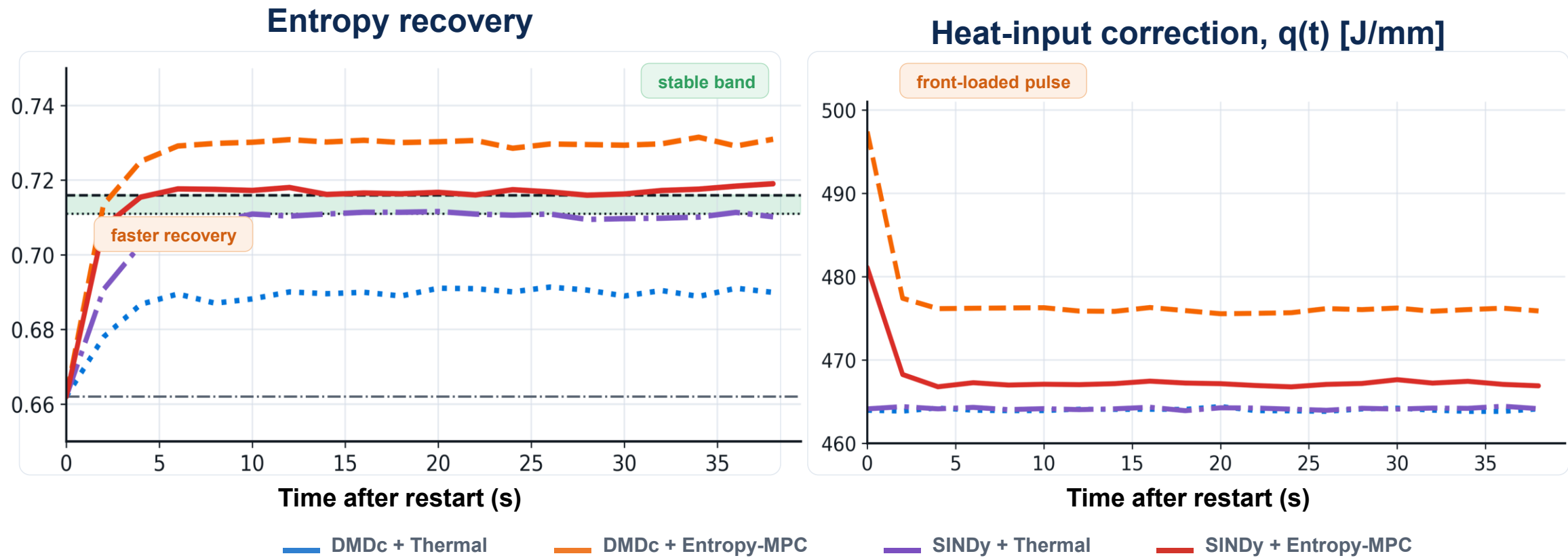
Entropy drop tracks flaw-risk behavior.



Arc sound provides an **in-situ quality-risk signal** before post-build inspection.

05 / DESIGN MARGIN & HEAT CORRECTION

Front-loaded heat correction accelerates restart recovery.

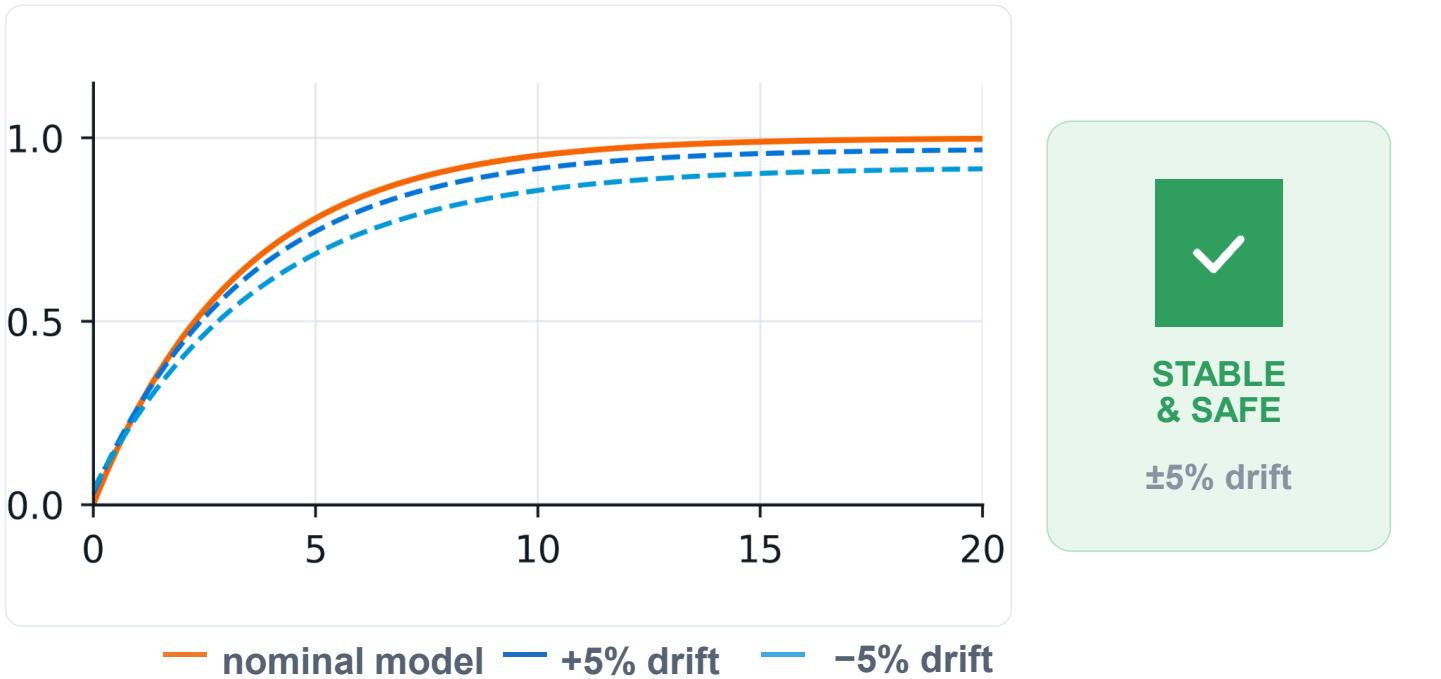


Entropy-aware MPC reaches the stable entropy region faster via a short front-loaded heat-input correction.

06 / ROBUSTNESS

Stable under $\pm 5\%$ model mismatch.

Recovery remains stable when **SINDy parameters** drift by $\pm 5\%$.



Design margin: 61.5% over thermal-only baseline.

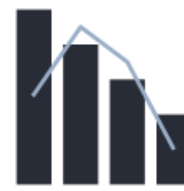
Results at
a Glance

KEY OUTCOMES



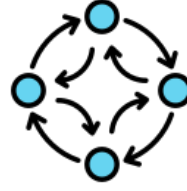
47%

Faster Restart Recovery



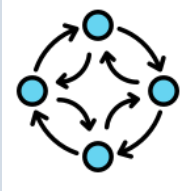
35%

Lower Quality-deviation AUC



$2.4e^{-4}$

P · Process Entropy → Flaw Proxy



-0.37

P · Pause-entropy Association



439 J/mm

Simulated Heat Pulse



60%

Surrogate Control Gain



TL/DR

Cold restarts can look electrically nominal while **acoustic entropy collapses**; entropy-aware MPC converts this hidden transient into a **short heat correction**, improving recovery by **47%** and reducing quality-deviation AUC by **35%**.

